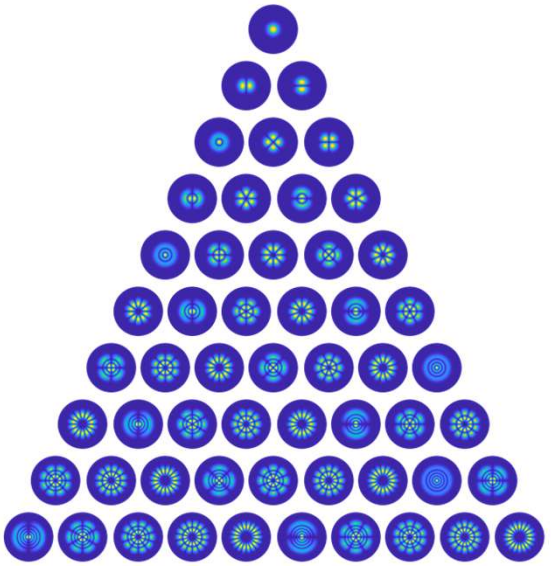
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 5



Prof. Dr.-Ing. Georg Rademacher

University of Stuttgart
Faculty 5: Computer Science, Electrical Engineering and Information Technology



25.-27. November 2025

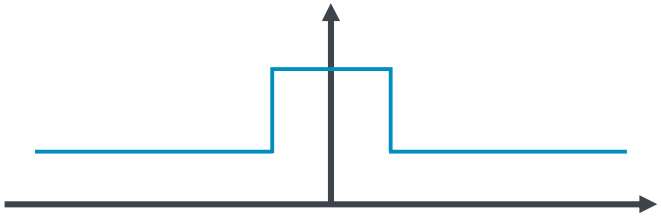
#InternationalDays Fakultät 5



- ✓ Entdecke Möglichkeiten für dein Auslandssemester!
- ✓ Lerne Partneruniversitäten kennen und spreche mit Austauschstudierenden!
- ✓ Feier mit uns die kulturelle Vielfalt unserer Fakultät - mit einem Programm von Studierenden für Studierende!

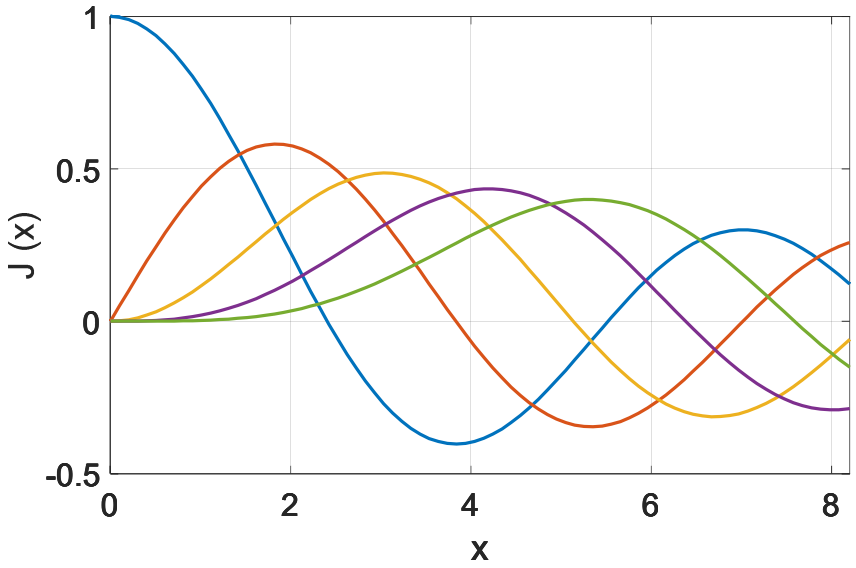
ISP International Service Point

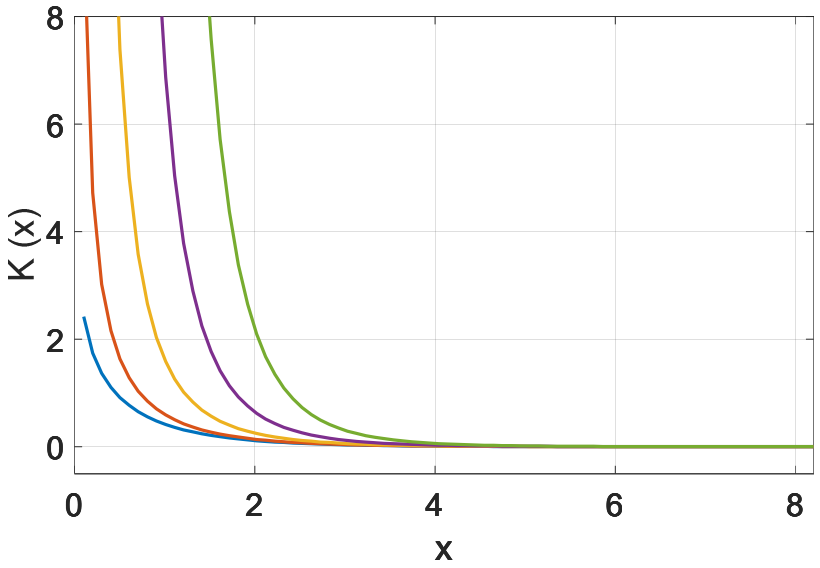
Linearly polarized (LP) waves



Besselfunctions

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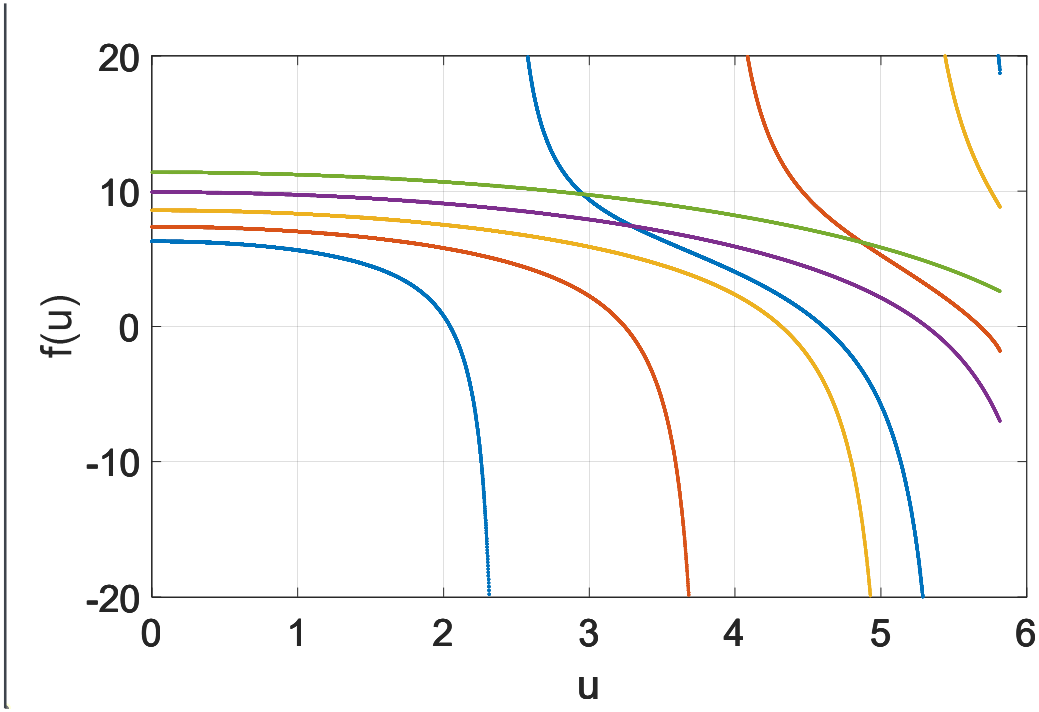




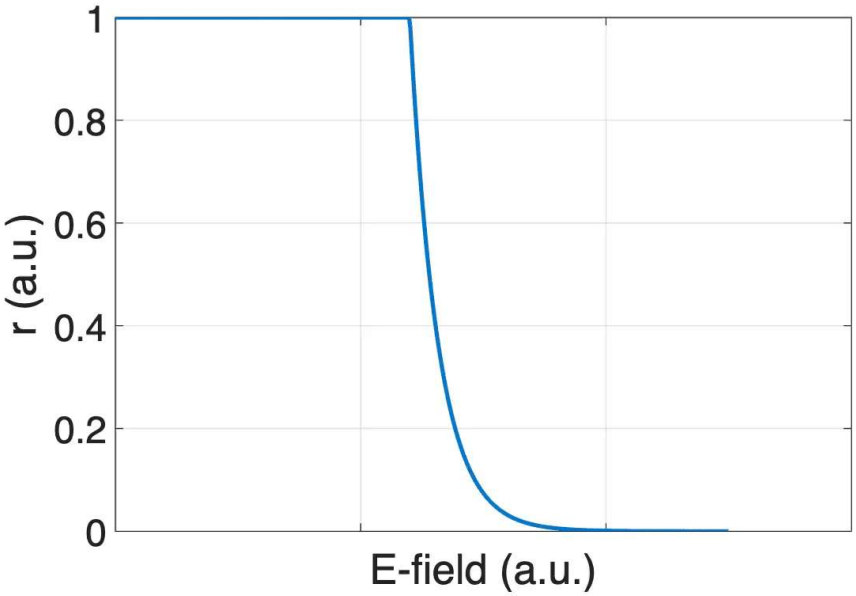
Linearly polarized (LP) waves

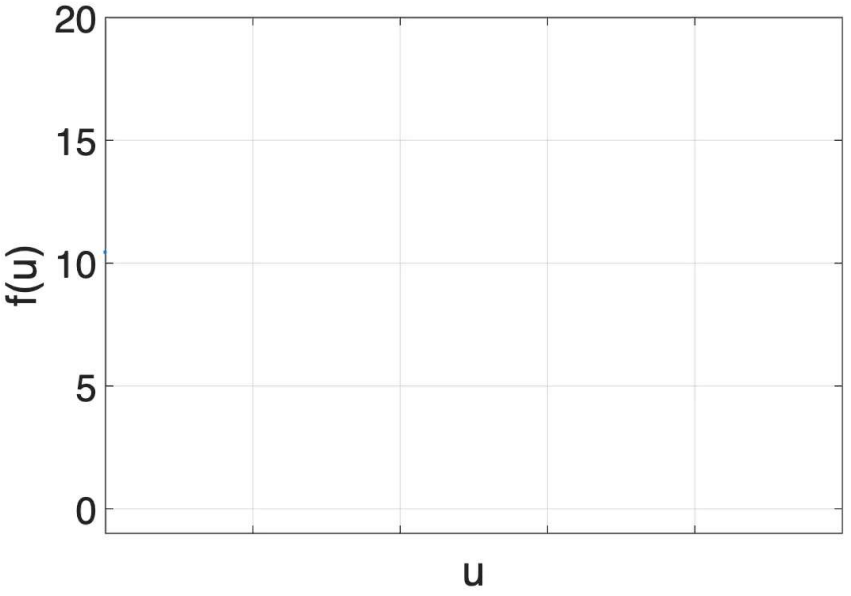
Linearly polarized (LP) waves

Linearly polarized (LP) waves



Linearly polarized (LP) waves

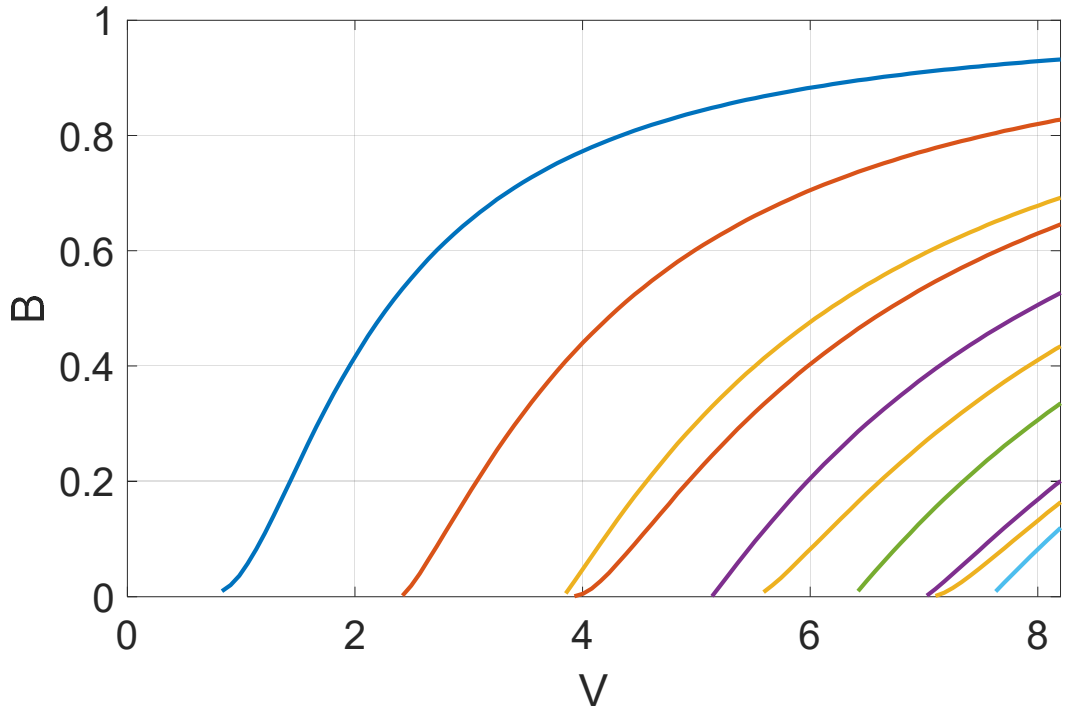




Linearly polarized (LP) waves

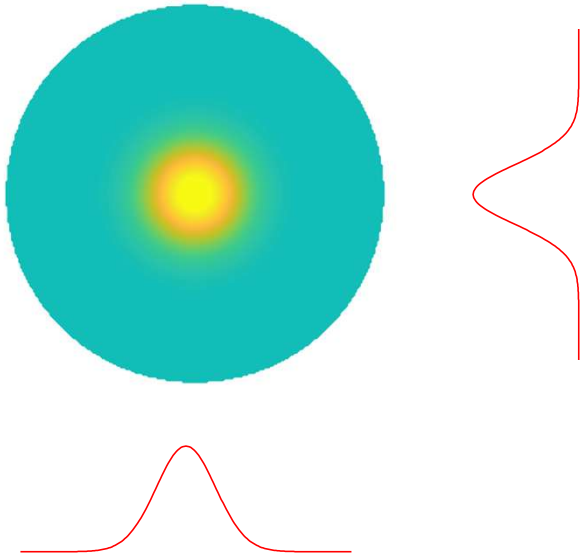
Linearly polarized (LP) waves

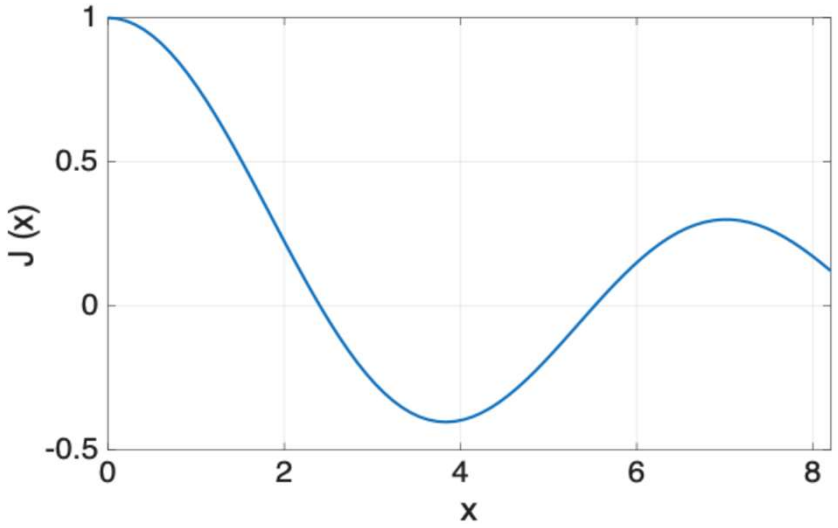
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Mode Fields in Step Index Fibers

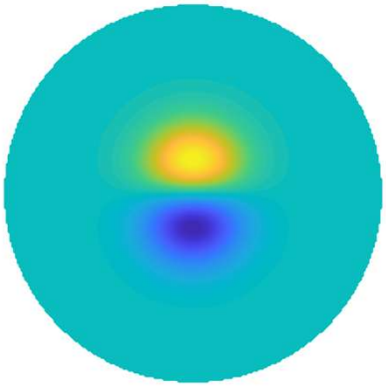
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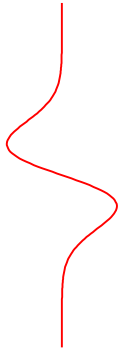


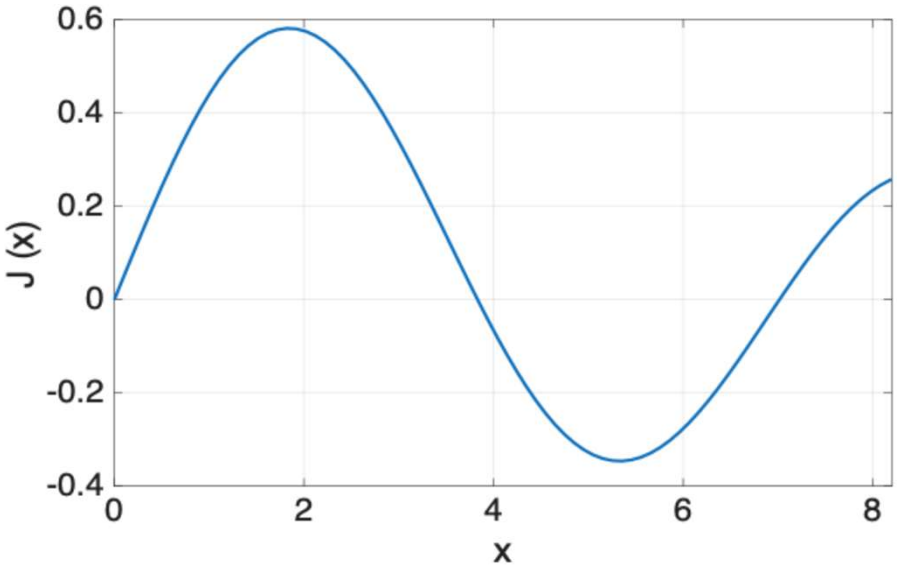


Mode Fields in Step Index Fibers

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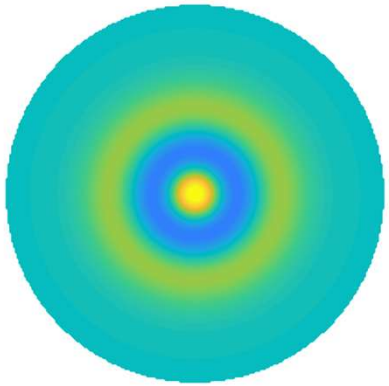


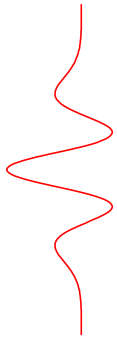


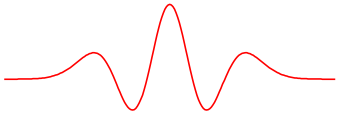


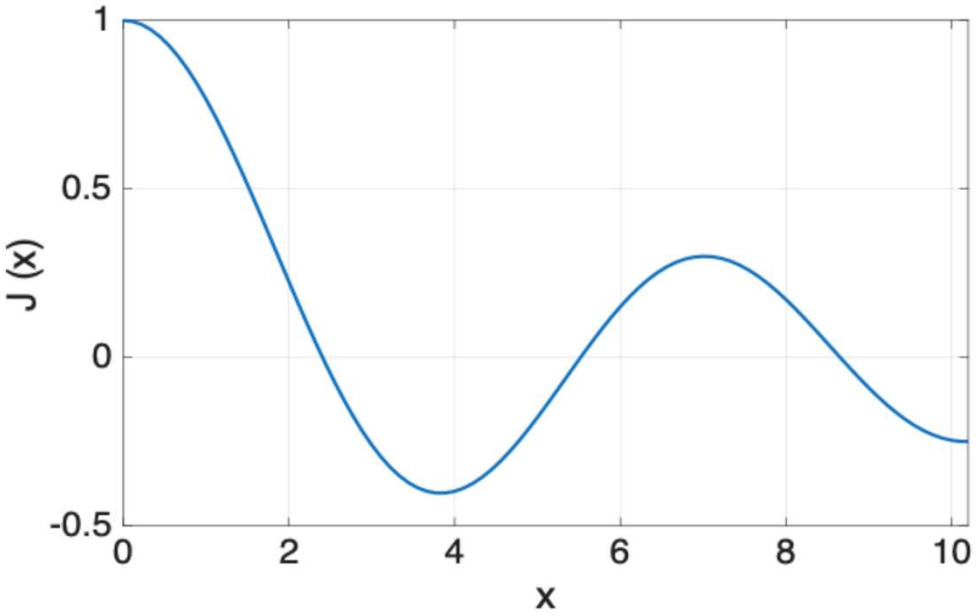
Mode Fields in Step Index Fibers

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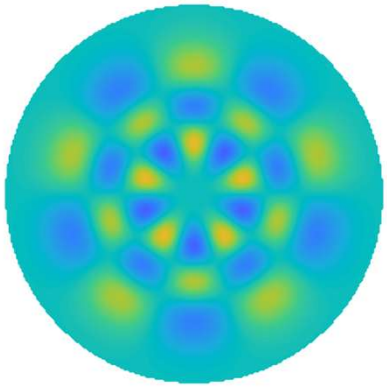


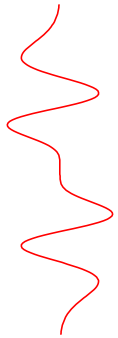


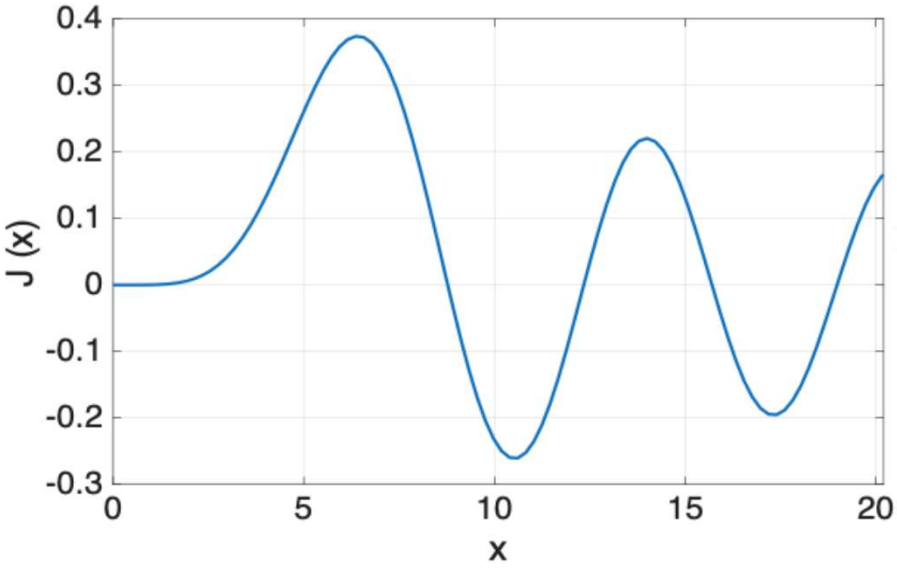


Mode Fields in Step Index Fibers

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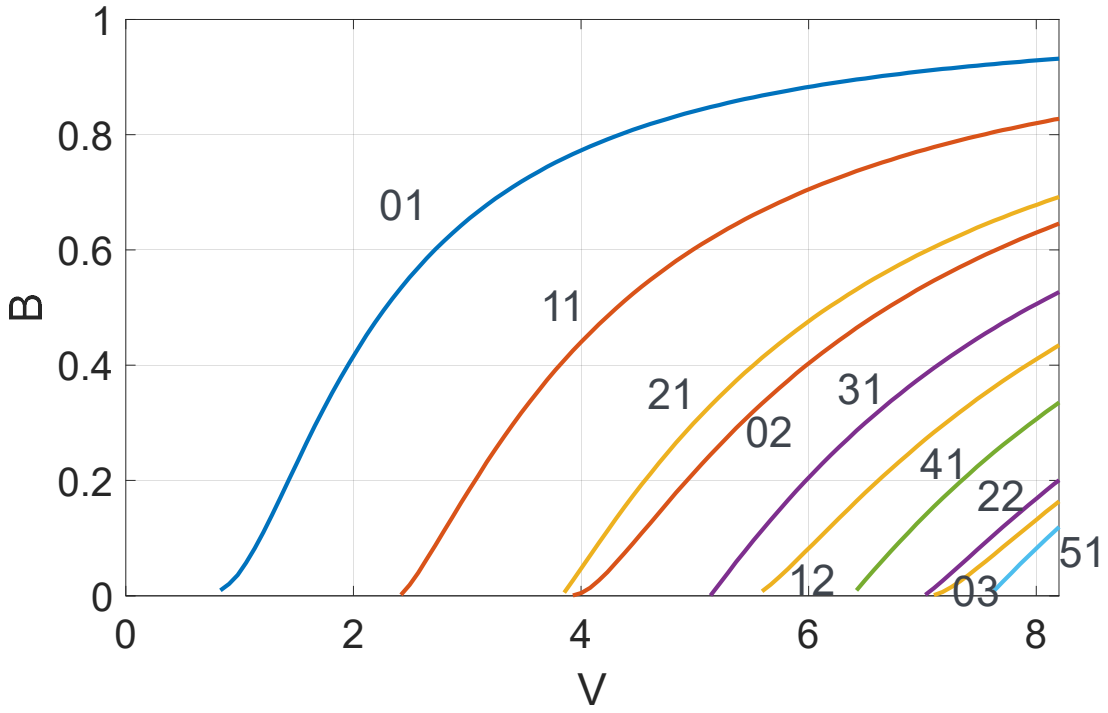




Single-mode fiber

Single-mode fiber

Linearly polarized (LP) waves

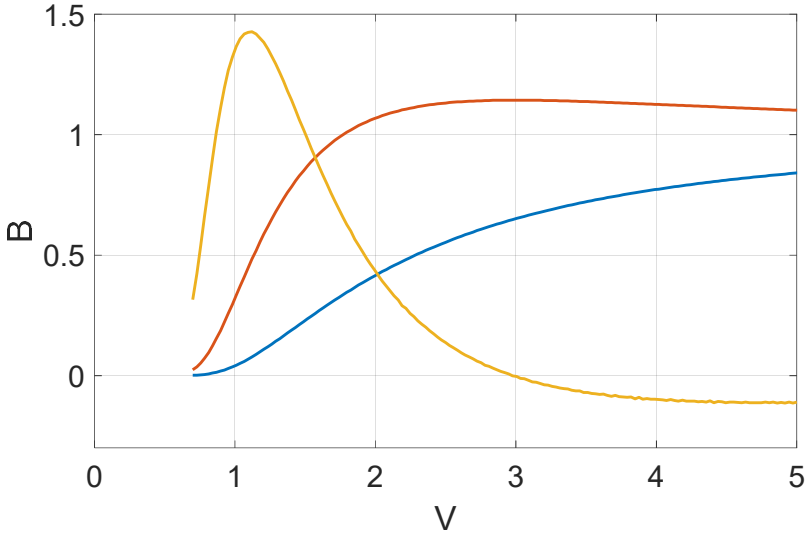


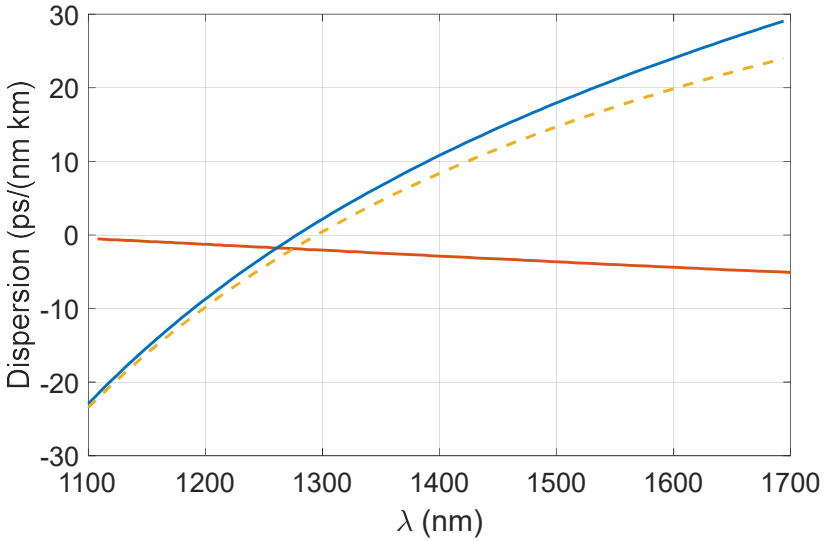
Chromatic dispersion

Chromatic dispersion

Chromatic dispersion of the fundamental mode

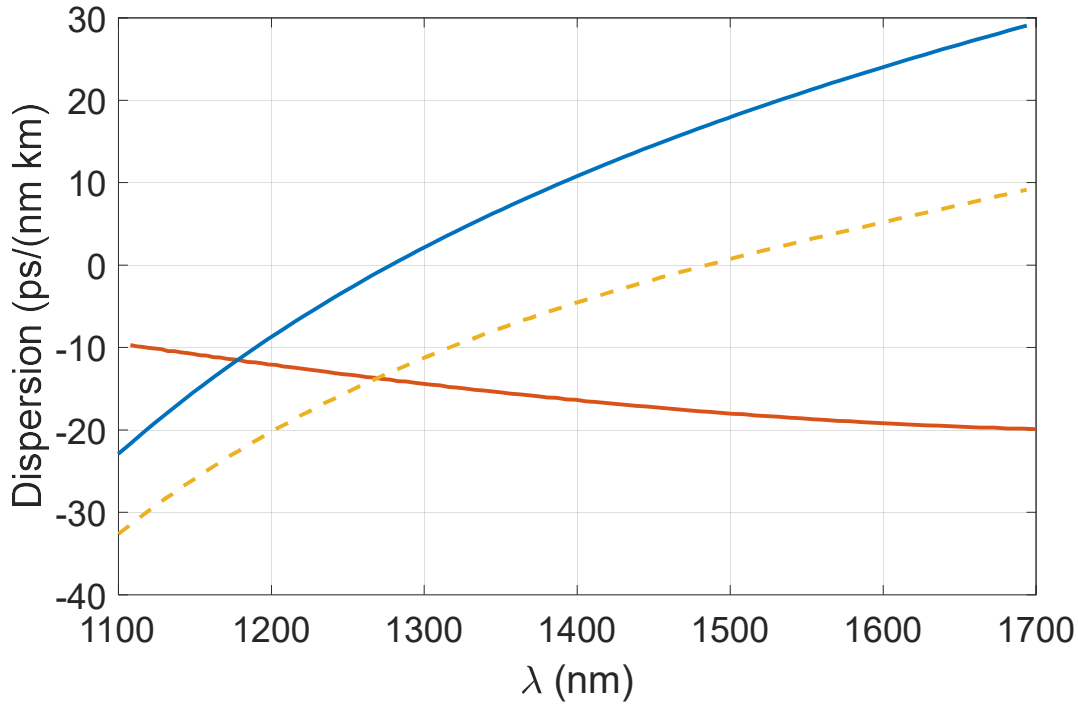
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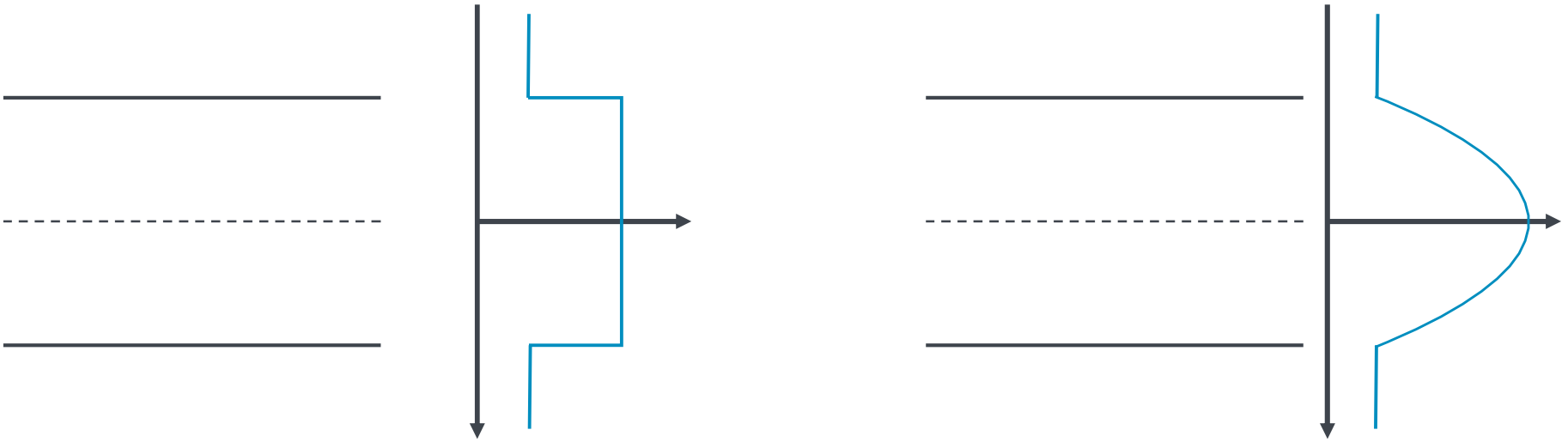
Chromatic dispersion of the fundamental mode

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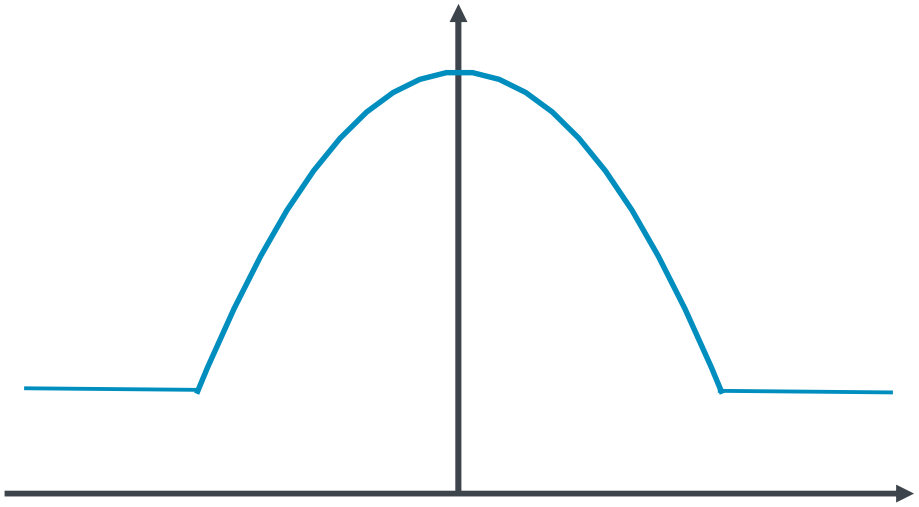


Graded-index fibers

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Graded-index fibers

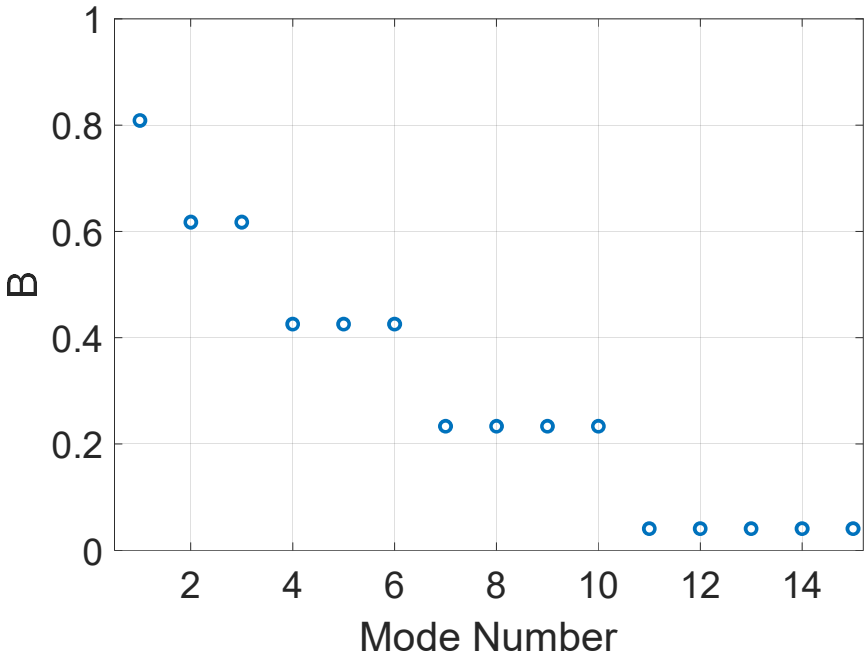


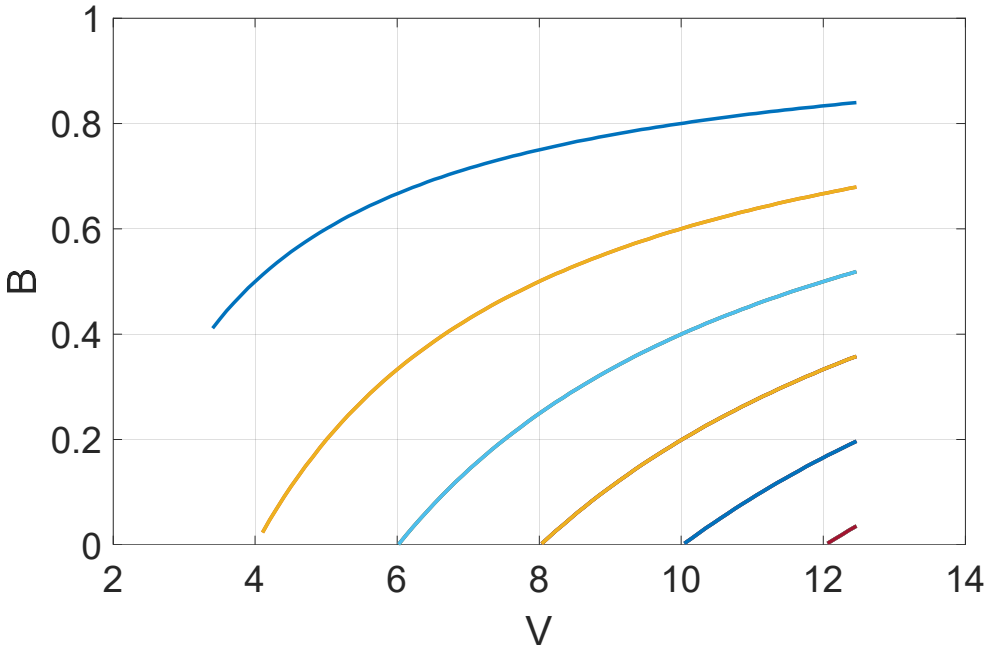
Graded-index fibers

Graded-index fibers

Graded-index fibers

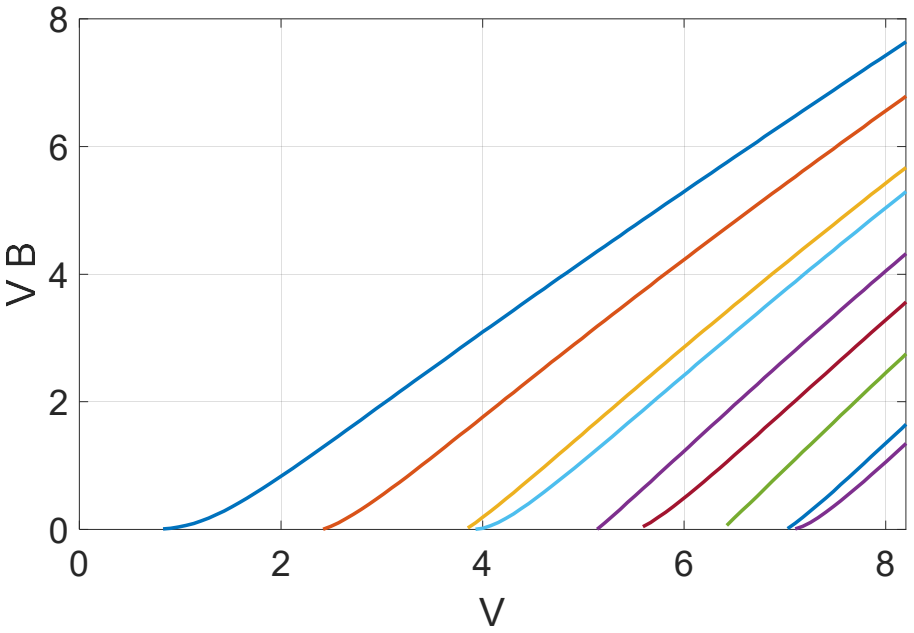
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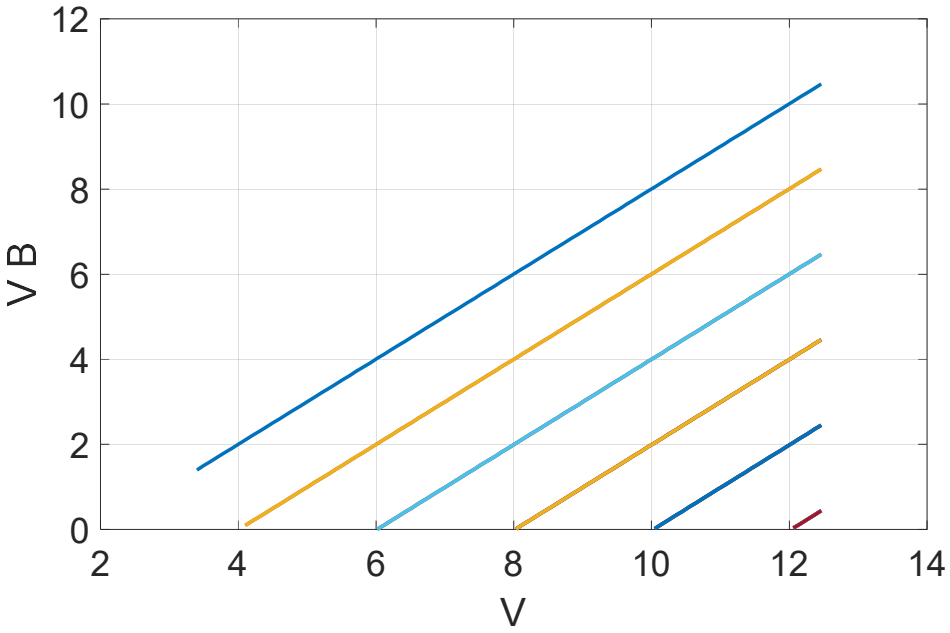




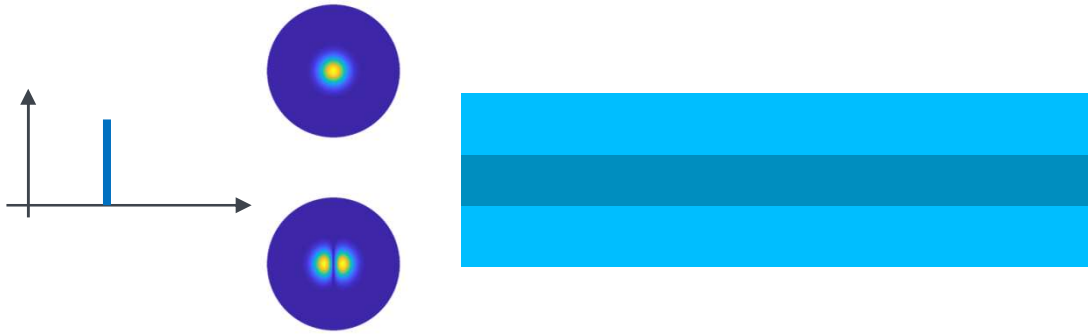
Graded-index fibers

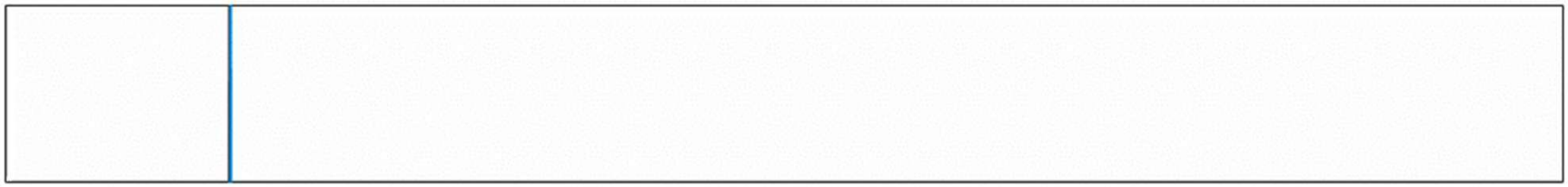
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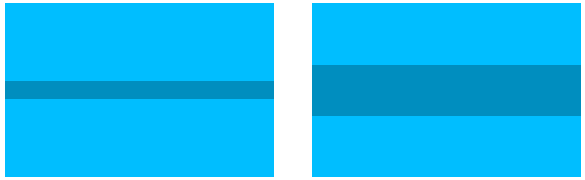


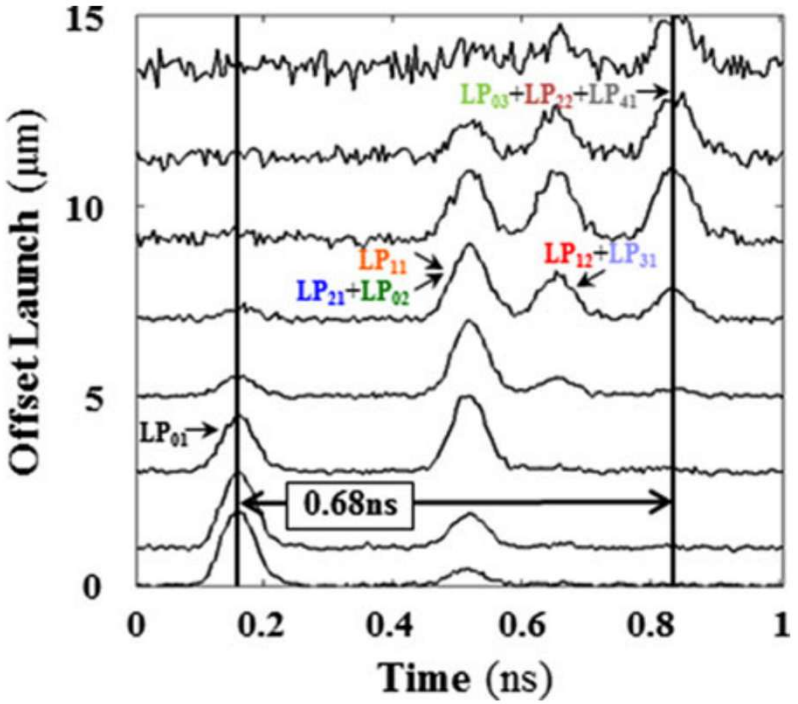
Graded-index fibers





Graded-index fibers

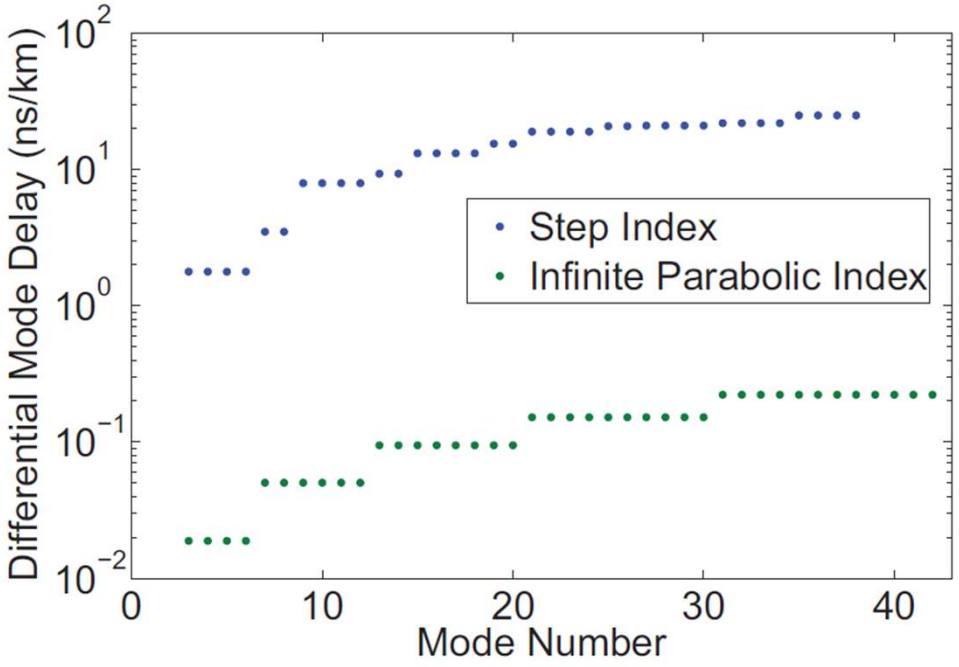




P. Sillard et al., Low-differential-mode-group-delay 9-LP-mode fiber, JLT 2016

Graded-index fibers

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$$\beta_{lp} = \cdot\left(2l + 4p - 2\right)^{1/2}$$

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Thank you!



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